

The Accuracy Illusion: How Overlapping Samples Manufacture Confidence, Not Skill, in Intraday Index Forecasting

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Abstract. Intraday forecasting systems routinely report directional hit-rates between 60% and 85%, and those figures move real capital. We show they are not valid estimates of skill. A k -bar-ahead forecast issued every bar shares $k-1$ of its k outcome bars with its neighbour; when the model's calls also persist, as they do on a live system we instrument, the reported hit-rate acquires a variance far larger than its nominal sample size implies. The reading is not biased upward. Its stated precision is. A forecaster with *exactly zero skill*, simulated against real NSE index bars, posts a $\geq 71\%$ session once in 48 while reporting an interval $1.68\times$ too narrow, rising to $3.40\times$ at the per-minute density these systems publish. We give the closed form behind this, the classical design effect with clusters given by runs of identical calls, and a calibrated interval built on it that covers at 93% where the conventional one covers at 73%. Applied to our own results it leaves nothing: across 48 model-index-horizon tests, 16 clear $p < 0.05$ scored per bar and *zero* clear it per event or at an effective sample size. A premium-selling strategy we had published as this project's one measurable edge fails the same correction. Break-even accuracy derived from measured move distributions is 71.5% at thirty minutes against a best achieved 51.1% on the same index, and at five minutes no accuracy suffices under a symmetric bracket. Code and raw results are public.

Keywords: Overlapping observations · Backtest overfitting · Directional forecasting · Negative results · Reproducibility · Index derivatives

1 Introduction

A system reporting 71% directional accuracy is, on its face, extraordinary. Work that controls for scoring unit, costs and multiple testing places realistic out-of-sample intraday index accuracy close to a coinflip [1,2], while surveys of the applied literature catalogue many higher figures reported without those controls [3].

This paper began as an attempt to build such a system. Over fourteen months we ran a live multi-horizon forecasting platform for Indian index derivatives, and

it reported 71%. That number was wrong. The session producing it cannot be re-audited, because the portion of the log holding it was lost before this analysis; every figure below comes from sessions that survive.

The econometrics of overlapping observations is not new [4,5], and Boudoukh et al. [6] show that overlap manufactures apparent long-horizon predictability which vanishes under correct inference. What that literature does not document is the magnitude of the distortion in a now-ubiquitous practice: the every-minute multi-horizon dashboard, where overlap meets a second ingredient, prediction persistence, and is amplified across a panel.

Contributions. (i) We measure the artefact as it occurs, showing that overlap alone changes nothing and that inflation requires overlap *and* persistent predictions, with persistence calibrated from a production log rather than assumed. (ii) We derive the effective sample size of such a stream, show it is governed by $\mathbb{E}[L^2]/\mathbb{E}[L]$ over run lengths, and calibrate an interval that covers at its nominal rate with no fitted constant. (iii) We apply it to our own results and to a complete control set — shuffled labels, injected positive controls, eight architectures, a block-bootstrap null — and report that nothing survives. (iv) We invert accuracy into a required-versus-achieved framing that shows the gap is not close, and is in places unbridgeable at any accuracy. Code, raw results and the free-data corpus are at <https://github.com/rachitranka25/accuracy-illusion>.

2 Related Work

Hansen and Hodrick [4] and Richardson and Smith [5] characterise inference under overlapping returns; Newey and West [7] give the standard heteroskedasticity- and autocorrelation-consistent correction. That work concerns regression coefficients. We apply the same logic to what practitioners publish, the session hit-rate, and reach the classical design effect [8,9] with clusters given by runs of identical calls. The contribution is its application to forecast streams, not the statistic.

On multiplicity, Harvey et al. [10] argue the conventional $t > 2$ threshold is indefensible after decades of factor mining. Two deflation statistics address the same concern from different directions: one discounts a reported Sharpe ratio by the number of trials behind it [11], the other estimates how often an in-sample winner fails out of sample [12]. Complementary controls come from White’s reality check [13], its refinement by Hansen [14], and the false-discovery-rate procedure of Benjamini and Hochberg [15]; Sullivan et al. [2] apply the reality check to a universe of technical trading rules, the direct precedent for our strategy sweep. On machine learning for finance, Fischer and Krauss [16] and Krauss et al. [17] report headline performance gross of costs, Gu et al. [18] give a comprehensive asset-pricing comparison with modest out-of-sample R^2 , and Zhang et al. [19] and Kolm et al. [20] demonstrate genuine short-horizon alpha — both from message-level exchange data rather than bars, the distinction that matters here.

Table 1 places this work against that set. The point is not that the cited studies are wrong but that the scoring unit differs: per-bar scoring of a multi-bar horizon, the condition under which the artefact arises, is characteristic of intraday dashboards rather than of the published work we cite.

Table 1. Methodological comparison with representative prior work. “Scoring unit” is the unit in which hit-rate or return is aggregated; per-bar scoring of a multi-bar horizon is the condition under which the artefact of Section 4 appears. MT: multiple-testing correction. NC: null control (shuffled labels or equivalent). C: transaction costs modelled. R: code and raw results public.

Study	Market	Granularity	Scoring unit	MT	NC	C	R
Fischer & Krauss [16]	S&P 500	Daily bars	Per day	—	—	pre-cost	—
Krauss et al. [17]	S&P 500	Daily bars	Per day	—	—	pre-cost	—
Gu et al. [18]	US equities	Monthly	Per month	partial	—	partial	partial
Zhang et al. [19]	LSE	Message-level LOB	Per event	—	—	—	partial
Kolm et al. [20]	Nasdaq	Message-level LOB	Per horizon	—	—	discussed	—
This work	NSE index	5-min bars, EOD chains	Per event	yes	yes	yes	yes

Multiple-testing control here means Bonferroni and Benjamini–Hochberg, which applied cleanly, plus the two deflation statistics, which we ran and found inapplicable and degenerate respectively (Section 6). We omit a row for “typical applied studies”: characterising that population would require citing specific papers as negative examples, and the argument does not need it.

3 Data and Method

All data is obtainable free from a retail brokerage account and public exchange archives: 26 265 and 26 267 five-minute bars on BANKNIFTY and NIFTY 50 from February 2025 to July 2026, 9.6k–14k bars on three further indices, 1 382 738 rows of end-of-day option chain with open interest, a ten-year chain history covering 2 181 days, 515 days of participant-wise open interest and 639 days of index futures carrying real volume. Two practical notes: index spot prints zero volume, so a feature computed as a “VWAP” on spot is an unweighted expanding mean and real volume exists only on futures; and the exchange’s *live* option-chain API is geo-restricted while its static end-of-day archive is not, which turned an assumed constraint into a decade of usable data.

Every accuracy below passes through one harness. Splits are expanding-window walk-forward with the k bars whose labels overlap the test window purged from training [21]. Accuracies carry Clopper–Pearson [22] intervals, exact binomial p -values and the Pesaran–Timmermann statistic [23], which is the appropriate null for directional accuracy. Every model is additionally trained on randomly permuted labels; if real and shuffled scores coincide, the pipeline is reading noise. A synthetic feature of known strength is injected as a positive control, since a pipeline that cannot recover planted signal cannot be trusted to report a null. Strategy returns use non-overlapping k -bar holds with explicit costs.

Units. All analysis is on 5-minute bars, so the 30-minute horizon is $k = 6$ and a session yields $n = 69$ forecasts. The production system emits a forecast every *minute*; pooled, its run lengths average 15.2 minutes with a longest of 263, and the pooled mean run in 5-minute bars is 3.4.

4 The Mechanism

4.1 Overlap is arithmetic

Measure the autocorrelation of the k -bar forward return two ways. On overlapping windows it reaches 0.841 at lag 1 and decays to approximately zero at *exactly* lag k ; sampled so that no two observations share a bar it is 0.067 at lag 1 and indistinguishable from zero thereafter. The pattern reproduces at $k = 6$ and $k = 12$, with the decay landing on lag 6 and lag 12 respectively. The correlation is a property of the sampling scheme, not of the market, and it is the whole of the mechanism below.

4.2 Persistence is the second ingredient

Overlap alone is insufficient, and getting this wrong was instructive: if predictions are drawn independently each bar, the hit indicator is i.i.d. Bernoulli regardless of how autocorrelated the outcome series is, and our first simulation made exactly that error and correctly found nothing. Real models do not re-decide every bar. Their features move slowly, so they repeat the same call for long stretches: measured on the production log at the 30-minute horizon, BANKNIFTY calls persist for a mean of 25.9 minutes and NIFTY 50 for 10.3, against a mean run of one for a model that re-drew each bar. When both the call and the outcome persist, one lucky trending stretch is recorded as dozens of hits.

4.3 What a zero-skill forecaster reports

We simulate a fair coin held for a run length drawn from the measured distribution, evaluated against real bars and scored as a dashboard scores it (Table 2).

Table 2. A forecaster with exactly zero skill, 30-minute horizon, scored per bar versus per event. BANKNIFTY; NIFTY 50 agrees within 0.6 points on every row.

	Per bar Per k bars	
Reported sample size per session	69	12
Stated binomial 95% CI	± 11.8 pts	± 28.3 pts
Realised spread across sessions	± 19.9 pts	± 28.0 pts
Understatement factor	1.68$\times$	0.99$\times$
$P(\text{session} \geq 60\%)$	16.2%	18.9%
$P(\text{session} \geq 71\%)$	2.1%	7.1%

The table states a point easily read backwards. Per-bar scoring does *not* produce extreme readings more often than honest scoring; it produces them *less* often. What it does is report them with an interval $1.68\times$ too narrow, because the quoted n of 69 is a fiction. Overlap manufactures unwarranted *confidence*, not high hit-rates, and that false precision is what misleads.

4.4 At the real granularity, and across a panel

The production system forecasts every minute, so a session yields about 313 scored forecasts rather than 69. Taking the outcome series from the log and simulating the same zero-skill forecaster, the understatement factor rises to $3.40\times$ while extreme sessions become slightly *rarer*, so every 5-minute-bar figure here understates the artefact. A dashboard also watches more than one number: two indices at four horizons is eight hit-rates per session, and simulating the panel jointly gives $P(\exists \text{series} \geq 71\%) = 17.1\%$ against 17.2% under independence — about one session in six, so a headline 71% arrives within the first weeks of live operation with no edge at all.

5 A Calibrated Estimator

5.1 Effective sample size

Let $h_t = \mathbf{1}\{\hat{y}_t = y_t\}$ and $\hat{p} = n^{-1} \sum_t h_t$. Practitioners quote $\text{Var}(\hat{p}) = p(1-p)/n$, valid only for independent h_t . In general

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n} \left[1 + 2 \sum_{j=1}^{n-1} \left(1 - \frac{j}{n}\right) \rho_j \right], \quad (1)$$

with $\rho_j = \text{Corr}(h_t, h_{t+j})$. Because the dependence is block-shaped rather than smoothly decaying, this collapses: if the sequence decomposes into runs of length L_i within which h_t is constant, then

$$n_{\text{eff}} = \frac{n}{\sum_i L_i^2/n} \xrightarrow{n \rightarrow \infty} \frac{n \mathbb{E}[L]}{\mathbb{E}[L^2]}. \quad (2)$$

This is the classical design effect [8,9]; what is new is that a forecast stream has exactly this structure and that nobody reports it. The inflation is governed by $\mathbb{E}[L^2]$, so the tail matters, not the mean.

5.2 Validation by coverage

Table 3 reports coverage against a forecaster of known skill whose calls persist per the measured distribution, with true skill measured from the generative process rather than assumed.

HAC and a block bootstrap (86.2%, omitted from the table) both under-cover by about nine points. The obvious diagnosis is bandwidth: the automatic

Table 3. Coverage of a nominal 95% interval on a session hit-rate. BANKNIFTY, 30-minute horizon, nominal $n = 69$; NIFTY 50 agrees within 2 points. HAC-L gives the Bartlett kernel a run-structure bandwidth; *derived* divides the design effect by the alternation factor of (3).

True skill	Naive	HAC	HAC-L	Design	Derived	n_{eff}
50%	73.4	86.2	88.0	99.7	93.2	11.9
54%	73.9	86.6	88.1	99.6	93.5	11.7
59%	73.0	85.2	86.8	99.6	93.5	10.9

Bartlett rule gives 3.7 bars at $n = 69$, close to the pooled mean run of 3.4, while the tail reaches 53. A run-structure bandwidth recovers one to two points and no more, so bandwidth was part of the problem and not the whole of it. The design effect (2) removes the failure mode entirely and over-corrects in the other direction, reaching 99.7%. That over-statement is derivable rather than incidental: treating runs as independent blocks ignores that a binary sequence’s runs alternate deterministically, a run of hits always being followed by a run of misses. For geometric run lengths of mean μ ,

$$\frac{\mathbb{E}[L^2]}{\mathbb{E}[L]} = 2\mu - 1, \quad \text{whereas exactly } \frac{1 + \rho}{1 - \rho} = \mu - 1, \quad (3)$$

so (2) over-states the dependence by a factor $(2\mu - 1)/(\mu - 1)$, a quantity that tends to 2 for long runs. Evaluated per session it averages 2.44 at a mean hit-run length of $\mu = 3.53$; the residual is the departure of real run lengths from geometry.

Dividing by that factor, computed from each sample’s own mean run length, introduces no free constant and gives 93.1–93.5% coverage across both indices and all three skill levels. We report it as the recommended estimator, and the raw design effect as the conservative bound.

5.3 Reliability, and a real session

Equation (2) depends on a tail-driven second moment estimated from a few dozen runs, and a tail is what a short sample estimates worst. Measured against a pooled reference of 10.6, a single session’s n_{eff} lands between $0.63\times$ and $1.73\times$ that value nine times in ten, so it should be estimated from a run of sessions and treated as an order of magnitude when it comes from one. That matters immediately. Applying the estimators to the production log, a BANKNIFTY session reporting 57.1% has a naive interval that excludes 50% and a corrected interval of $[34.4, 79.9]$ that does not. The best NIFTY 50 session, 69.6% over 335 forecasts, has a corrected interval of $[50.0, 89.1]$, which does exclude 50% — but at the 5th percentile of the n_{eff} uncertainty it becomes $[45.1, 94.0]$ and no longer does. The margin is smaller than the estimator’s own noise, so the session supports no claim of skill. An earlier draft reported that it did.

Per-event scoring is unbiased, needs no estimator, and is what we use for every backtest here. It is nonetheless insufficient for a live stream: a dashboard

cannot wait k bars or discard forecasts it has already shown; subsampling throws away 83% of the observations; and which of the k offsets is chosen moves the same session by 7 to 20 points, so an analyst free to pick the offset is free to pick the answer.

6 Predictability Under Correct Scoring

6.1 Is there anything to predict?

Before asking which model is best, we ask whether any model could succeed. An unconstrained forest was fitted until it memorised the training data, on five indices across two exchanges (Table 4). Each memorises completely and generalises at chance; permuting labels moves accuracy by 0.3 to 0.7 points; no real-label result is distinguishable from a coinflip ($p = 0.22$ to 0.91); and both injected positive controls are recovered at $p < 0.001$. The null is a property of the data, not of the estimator.

Table 4. Model-independent ceiling test, 30-minute horizon. Training accuracy is 100.0% on both rows. FINNIFTY, MIDCPNIFTY and SENSEX were run identically and agree within 0.8 points on every column (real 49.7–50.2, shuffled 50.1–50.8). The positive controls have recoverable accuracies of about 55% and 65% by construction.

Index	n Real Shuffled Majority $s=0.10$ $s=0.30$					
BANKNIFTY	16 856	50.5	50.2	49.3	52.4	63.5
NIFTY 50	16 853	50.0	50.2	49.8	52.0	63.7

6.2 Eight model families, three horizons

Eight families were trained on identical splits, features, seeds and controls: logistic regression, gradient boosting, LightGBM [24], XGBoost [25], k -nearest neighbours, an LSTM [26], a Transformer [27] and a compact Temporal Fusion Transformer [28], each scored three ways. Counting families significant at $p < 0.05$ per horizon and index, per-bar scoring yields 0 and 2 at 15 minutes on BANKNIFTY and NIFTY 50, 3 and 1 at 30 minutes, and 5 and 5 at 60 minutes: 16 of 48 tests. At an effective sample size, and again scored one observation per event, the count is *zero* in every one of the six cells.

Scored per bar, XGBoost reaches 51.3% on BANKNIFTY at $p = 0.001$. That was, in an earlier draft, our one surviving positive result, described as robust to Bonferroni correction. It does not survive our own correction: re-scored one observation per 6 bars it is 50.4% at $p = 0.692$, and widening intervals to n_{eff} puts 50% inside every one. Across all 48 tests the count falls from 16 to zero. The count of *apparent* discoveries rises with horizon, from none at 15 minutes on BANKNIFTY to five at 60 on both indices, exactly as it should if the artefact scales with k ; the count of surviving discoveries stays at zero.

The zoo also contains its own diagnostic. The NIFTY 50 Transformer sits at 48.4% with $p < 0.001$, which at the nominal n is 4.3 standard errors *below* chance, where the largest $|z|$ across 48 tests should land near 2.9–3.2. That is not a broken model but the variance inflation of Section 4 surfacing in our own results; at n_{eff} it vanishes.

6.3 Eighty-six strategies

Separately, 86 parameterised rules — momentum, mean reversion, moving-average crosses, RSI, Bollinger bands, breakout, gap, volatility regime, time-of-day and streak variants — were evaluated on non-overlapping 30-minute holds at 4 bp round-trip cost with a one-sided test for profit. Two have a positive Sharpe ratio; none is significantly profitable, before or after Benjamini–Hochberg.

The deflated Sharpe ratio returns an implausible null benchmark of 7.77 here, because trial Sharpe dispersion is dominated by deterministic turnover differences rather than by noise, violating the assumption it rests on. We replaced it with a circular block bootstrap [29] that resamples forward returns while holding each strategy’s positions fixed. The best observed Sharpe is 0.21 against a null best-of-86 median of 0.44, with $P(\text{null} \geq \text{observed}) = 0.658$: the winner of the sweep is worse than what selection alone produces from no edge at all. The probability of backtest overfitting via combinatorially symmetric cross-validation [12] is degenerate here and is reported as such.

7 Economic Significance

A hit-rate without a payoff ratio is not a result. For a symmetric bracket with target and stop at the typical move m and round-trip cost c , expectancy is $m(2p - 1) - c$, so break-even requires

$$p^* = \frac{1}{2} + \frac{c}{2m}. \quad (4)$$

The ratio c/m is the entire problem (Table 5).

Two consequences follow. At the shortest horizon the median move is smaller than the round trip, so under a symmetric bracket p^* exceeds 100% and no accuracy suffices. That conclusion is specific to the bracket: moves are right-skewed, with a mean-to-median ratio near 1.4 at every horizon, so a hold-to-horizon rule requires 86% on BANKNIFTY and 93% on NIFTY 50 instead — not impossible, merely far beyond anything reported here or in the literature we cite. The point survives either payoff model.

Second, instrument choice can make the requirement unreachable. On the BANKNIFTY 30-minute horizon, where the median move is 9.3 bp, break-even is 55.4% at 1 bp of cost, 60.8% at 2 bp, 71.5% at 4 bp and 93.0% at 8 bp; at the 20 bp that a retail at-the-money option round trip costs over that horizon, $c > m$ and $p^* > 1$, so a perfect forecaster still loses on the median trade. The

Table 5. Break-even accuracy at 4 bp round-trip cost. p_{med}^* assumes a symmetric bracket at the median move; p_{mean}^* is the requirement for a hold-to-horizon rule with no bracket. Achieved is the best per-event accuracy of the eight families. Moves are in index points.

Horizon	Median	Mean	p_{med}^*	p_{mean}^*	Achieved
<i>BANKNIFTY</i>					
5 min	22.1	30.8	>100	86.4	—
15 min	37.7	52.9	79.8	71.3	51.3
30 min	52.3	73.8	71.5	65.3	51.1
60 min	72.9	103.8	65.4	60.8	52.2
<i>NIFTY 50</i>					
5 min	8.1	11.5	>100	93.0	—
15 min	14.1	19.8	84.9	74.9	50.5
30 min	19.7	27.8	75.1	67.8	51.2
60 min	27.7	38.5	67.7	62.7	50.1

instrument most retail participants actually use, over the horizon they actually hold it, has no directional accuracy that makes it profitable.

A genuine 85% would therefore clear break-even by 13.5 points on BANKNIFTY futures and still remain unprofitable on options: such a claim is either an artefact of the kind measured in Section 4 or true and far less valuable than it sounds.

8 What Is Predictable Instead

These results went through the same harness, and what it leaves is narrow enough to state precisely: *none of this project’s findings survives as a finding of ours.*

Volatility. A binary magnitude target reaches 70.8% per bar and 71.3% per event on BANKNIFTY (72.0 and 72.1 on NIFTY 50), with shuffled controls at 52–54%. That it is untouched by the correction which destroyed the directional result matters, since an instrument flattening everything would be useless. But a persistence rule already scores 67.4%, the model’s edge is +3.3 to +3.7 points per event, and at n_{eff} the model and baseline intervals overlap. This confirms volatility clustering [30,31] rather than discovering anything.

Variance risk premium, withdrawn. Short at-the-money straddles held to expiry, on real end-of-day premiums, across four entry points: no configuration is significantly profitable on either index, before or after Benjamini–Hochberg, and no bootstrap interval on the mean excludes zero on the profitable side. Recentring each configuration and resampling gives a best-of-four null median of 17.8 against an observed 12.1 on NIFTY ($P = 0.68$). The exchange also moved BANKNIFTY options from weekly to monthly expiry mid-sample, so the indices were never comparable: split on expiry spacing, the weekly sub-sample returns −67.5 points per trade and the monthly +57.5. The premium is real and documented [32,33]; our measurement of it as an edge is withdrawn.

Option-chain positioning. Over 2181 days, after regressing out the price component, six of eight residual features clear $p < 0.05$ uncorrected and *one* survives Benjamini–Hochberg; the daily change in the put–call open-interest ratio ($IC = 0.059$, $t \approx 2.8$, $\approx 51\%$ hit-rate), also the only feature strengthening in the second half. Flow beats position. Nothing in the price-only set clears $p < 0.05$ at all.

9 Limitations

The alternation correction assumes geometric run lengths; real runs depart from that, which is why coverage lands near 93% rather than 95%, and its input is imprecisely measured from a single session. Two indices carry the primary intraday analysis over seventeen months in one market; the ceiling test replicates across five indices and two exchanges, but the model zoo and strategy sweep do not. Break-even is computed for two payoff models, neither of which models the price path. Two figures that were re-run did not reproduce and are withdrawn in place rather than quietly deleted. Our Temporal Fusion Transformer is a compact variant, not the published architecture [28]. Finally, the order-flow hypothesis, where the literature locates genuine short-horizon alpha [20,19], is untested here for want of message-level history.

10 Conclusion

The headline metric of intraday directional forecasting is not merely noisy; its stated precision is systematically wrong, by enough to make a coinflip look like a discovery. A zero-skill forecaster posts a $\geq 71\%$ session once in 48, and once every six sessions somewhere on a modest panel, while reporting an interval $1.68\times$ too narrow at 5-minute granularity and $3.40\times$ too narrow at the density these systems publish.

Applied to our own work the correction leaves nothing standing. Sixteen of 48 model–index–horizon tests clear $p < 0.05$ per bar and none clears it per event; 86 strategies produce a winner worse than a bootstrap null; premium selling, published as this project’s one measurable edge, fails the same correction; and the one untouched result confirms volatility clustering rather than finding anything new.

The honest summary is that this project found nothing tradeable, and its value lies in how thoroughly that was established. What we offer is a method: an effective sample size for overlapping persistent forecasts, a calibrated interval built on it, and a framing that turns an accuracy figure into a question about cost. Reporting practice should change to match, and the change is cheap. Scoring unit, effective sample size, multiplicity correction, a shuffled-label control and an explicit cost model are not optional refinements. Without them an accuracy figure does not mean anything, and we know that because we published several that did not.

Reproducibility, data availability and scope. Every number here is regenerated by a script in the public repository, with the raw output of each committed alongside it, at <https://github.com/rachitranka25/accuracy-illusion>. That repository holds the project’s complete research record, including the positive results withdrawn above; all experiments use a fixed seed, and collectors for every data source are included so the corpus can be rebuilt. This is a measurement study, not investment advice: its central finding is that the reported accuracy of intraday directional systems is not a valid estimate of skill, a conclusion we apply first and most severely to our own published claims.

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